

# CHAPTER 1

## INTRODUCTION

Music plays a vital role in human life and is deeply connected to our emotions, experiences, and psychological well-being. In the modern digital era, millions of songs are accessible through online streaming platforms such as Spotify, YouTube Music, and Apple Music. Although these platforms provide personalized recommendations, most traditional recommendation systems rely on users' historical listening behavior, genre interests, ratings, and collaborative filtering techniques. Such systems fail to consider the emotional state of the user at the moment of listening, which is a critical factor influencing music preferences.

Human emotions are dynamic and often change rapidly based on internal thoughts or external circumstances. Listening to music that aligns with or enhances one's emotional state can help individuals relax, improve concentration, reduce anxiety, or elevate mood. Therefore, there is a growing need for a more intelligent and human-centered recommendation system that understands user emotions and suggests music accordingly.

The Emotion-Driven Music Recommender System aims to bridge this gap by incorporating emotion recognition capabilities into music recommendation technology. The system analyzes real-time emotional cues of the user through facial expressions or speech/sentiment signals using machine learning and deep learning techniques. Modern models such as Convolutional Neural Networks (CNN), LSTM networks, and advanced sentiment analysis algorithms enable accurate emotion classification into categories such as happiness, sadness, anger, fear, neutral, and surprise. Once the emotional state is detected, the system maps the identified emotion to suitable music genres or playlists using an emotion-music correlation model.

This approach introduces a more empathetic and interactive music experience compared to conventional methods. It allows music services to adapt to users' mental states, thus enhancing engagement, personalization, and user satisfaction. The system can be implemented as a mobile or desktop application and has potential applications in entertainment, health care, therapy, education, and wellness-based environments.

## 1.1 Objectives

- **To detect and analyze the real-time emotional state of the user** using facial expression recognition and/or speech-based sentiment analysis techniques.
- **To recommend personalized music based on the detected emotions** by mapping emotional states to suitable music genres, playlists, or song types.
- **To develop a recommendation engine** that integrates content-based filtering and emotion-based mapping to improve recommendation accuracy and relevance.
- **To create an interactive and user-friendly interface** that allows smooth interaction between the user and the system through image, video, or audio input.
- **To construct a music database / integrate streaming APIs** (such as Spotify API or YouTube Music API) for dynamic and scalable song retrieval.

## 1.2 Problem Statement

Traditional music recommendation systems rely mainly on user listening history, ratings, and collaborative filtering, but they do not consider the user's real-time emotional state. Since emotions significantly influence music preferences, users often struggle to find songs that match their mood at a given moment. Therefore, there is a need for an intelligent system that can detect the user's current emotion using facial expression recognition or sentiment analysis and automatically recommend suitable music, providing a more personalized and emotionally responsive listening experience

## CHAPTER 2

### LITERATURE SURVEY

Emotion-aware recommendation systems have gained increasing attention over the past decade due to advances in artificial intelligence, machine learning, and human-computer interaction.

#### 1. Facial Expression-Based Emotion Recognition (2020)

A study by researchers utilized **Convolutional Neural Networks (CNNs)** to detect facial emotions from real-time images. The model was trained on FER-2013 and CK+ datasets to classify basic emotional states such as happy, sad, angry, fear, neutral, and surprise. The study demonstrated that deep learning models significantly improve accuracy in emotion classification compared to traditional machine-learning algorithms.

#### 2. Speech-Based Emotion Classification (2019)

Another work explored the use of **speech signal processing and MFCC (Mel-Frequency Cepstral Coefficients)** combined with **LSTM neural networks** for detecting emotional tone in voice recordings. Results proved that audio-based sentiment recognition is effective for applications where facial inputs may not be available, providing another modality for emotion-centered recommendation systems.

#### 3. Music Recommendation Using Sentiment-Based Filtering (2021)

A research project implemented an emotion-driven recommendation framework that mapped emotional states to music metadata using **content-based filtering**. The system extracted features such as tempo, rhythm, energy level, and lyrics sentiment to align music with emotional categories. Experimental evaluation showed improved user satisfaction compared to popularity-based recommendations.

#### **4. Hybrid Recommender Systems for Personalized Music (2022)**

A hybrid model combining **collaborative filtering and emotional context modeling** was proposed to overcome the cold-start problem. The system integrated users' emotional data with similarity-based recommendation algorithms to produce more accurate playlists. The evaluation demonstrated reduced search time and improved personalization.

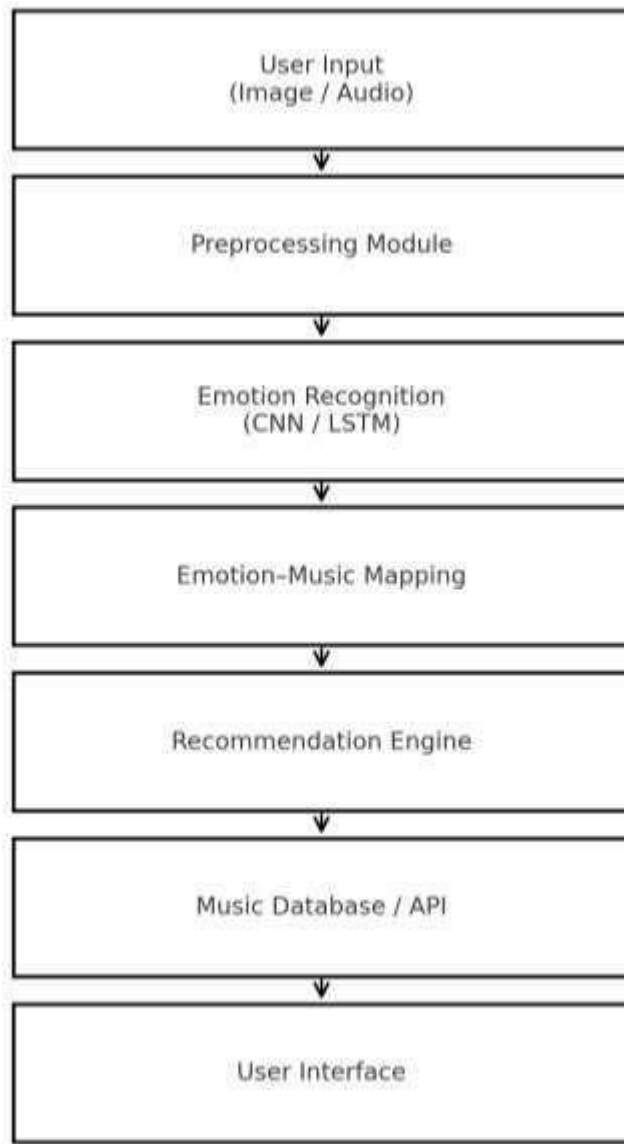
#### **5. Emotion-Driven Multimedia Applications for Mental Health (2023)**

Recent research has addressed the therapeutic role of emotion-aware systems. By utilizing AI models to detect stress or sadness and suggesting relaxing or motivational music, emotion-based recommenders proved beneficial for mental wellness and mood regulation. The study highlighted practical real-world applications such as stress-relief apps, healthcare systems, and wellness platforms.

## CHAPTER 3

### SYSTEM DESIGN

#### 3.1 ARCHITECTURE



**Figure 3.1: Block Diagram of Facial Emotion Recognition System.**

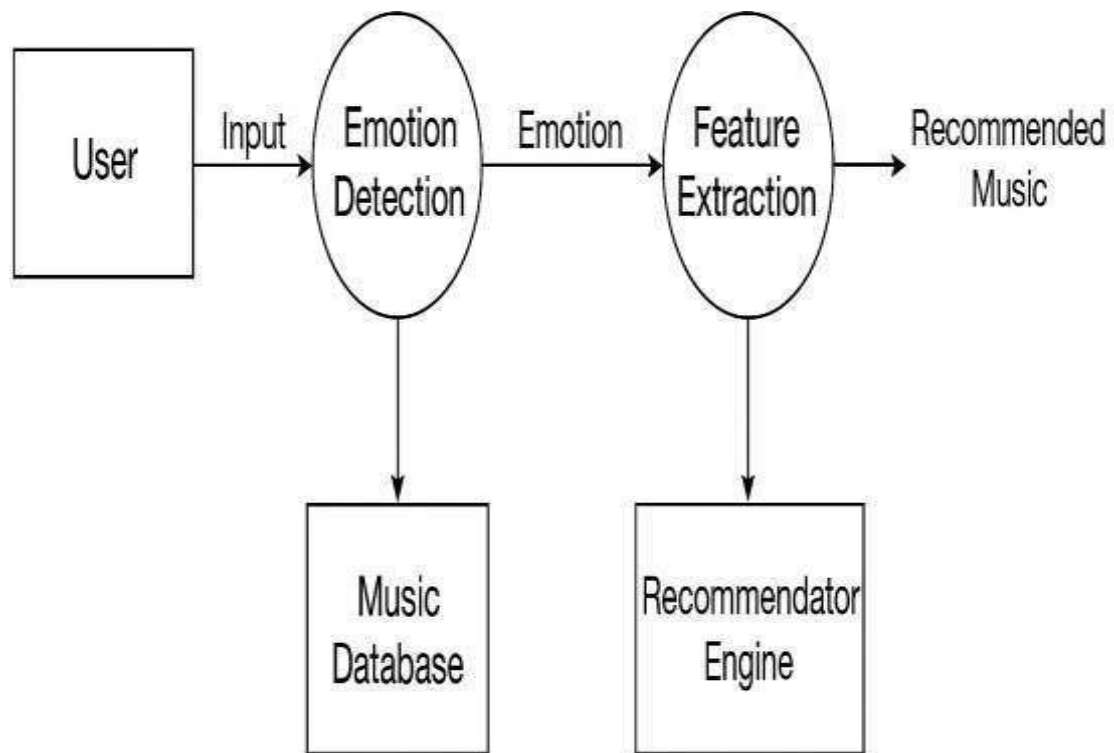
**3.2 DATAFLOW DIAGRAM**

Figure 3.2: Data flow Diagram of Facial Emotion Recognition System.

## CHAPTER 4

### METHODOLOGY

The methodology describes the step-by-step process followed to design, develop, and implement the Emotion-Driven Music Recommender System. The system is designed to analyze the emotional state of the user and recommend personalized songs accordingly. The methodology consists of the following major stages:

#### 1. Data Collection

- Collect image datasets (facial emotion images) from publicly available repositories such as FER-2013, CK+, JAFFE or live webcam inputs.
- Collect music datasets containing audio files along with metadata (mood labels, genre, tempo, lyrics).
- Preprocess images to uniform size, grayscale conversion, and normalization.
- Preprocess audio files to extract required features like MFCC, tempo, pitch, and spectral contrast.

#### 2. Emotion Detection

- Use deep learning-based **CNN (Convolutional Neural Network)** or **transfer learning models such as VGG16/ResNet** to classify facial expressions.
- Classes used: **Happy, Sad, Angry, Neutral, Fear, Surprise, Disgust**
- Train model using labeled images and validate accuracy.
- Output of the model: Predicted emotion label.

#### 3. Feature Extraction

- Extract high-level features from music datasets including:
  - Audio features (MFCCs, Chroma, Rhythm)
  - Metadata (artist, genre, release year, rating)
  - Lyrics sentiment using NLP-based models (optional)
- Convert extracted features into feature vectors to be used for matching.

## 4. Emotion-to-Music Mapping

- Create a mapping rule or model to associate each detected emotion with corresponding music features.
- Example mapping:

### **Emotion Suggested Songs Characteristics**

|         |                                                 |
|---------|-------------------------------------------------|
| Happy   | Fast tempo, high energy, pop/dance              |
| Sad     | Slow tempo, soft acoustic, piano                |
| Angry   | Rock/rap, high bass intensity                   |
| Neutral | Mixed or personalized history-based suggestions |

## 5. Recommendation Engine

- Apply content-based filtering or hybrid recommendation techniques.
- Calculate similarity score based on cosine similarity or Euclidean distance between user emotion vector and song feature vector.
- Retrieve top-N most relevant songs.
- Present playlist to UI interface.

## 6. System Integration

- Create GUI/mobile/web interface for user interaction.
- Capture user image and perform real-time detection.
- Display recommended playlist based on detected emotional state.
- Allow user to play, skip, rate and improve recommendations.

## 7. Evaluation & Performance Testing

- Validate emotion detection accuracy using test dataset and confusion matrix.
- Evaluate recommendation accuracy using precision, recall, MAE/RMSE.
- Conduct user satisfaction surveys to measure real-world performance



## ALGORITHM

Input: User input image, Music Database

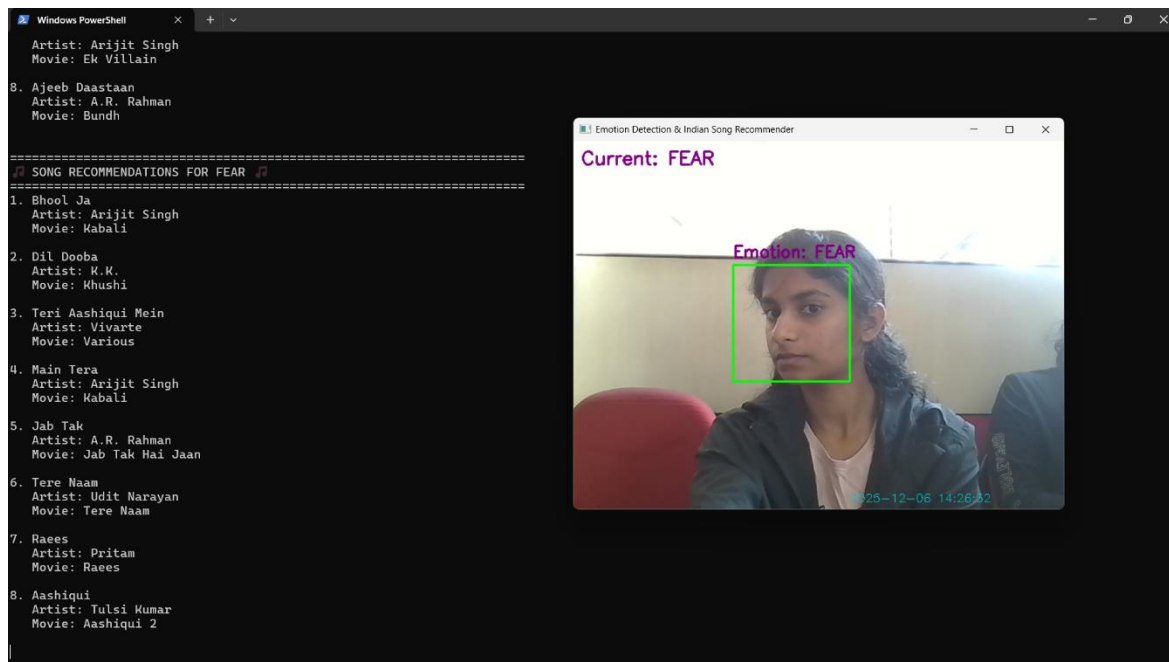
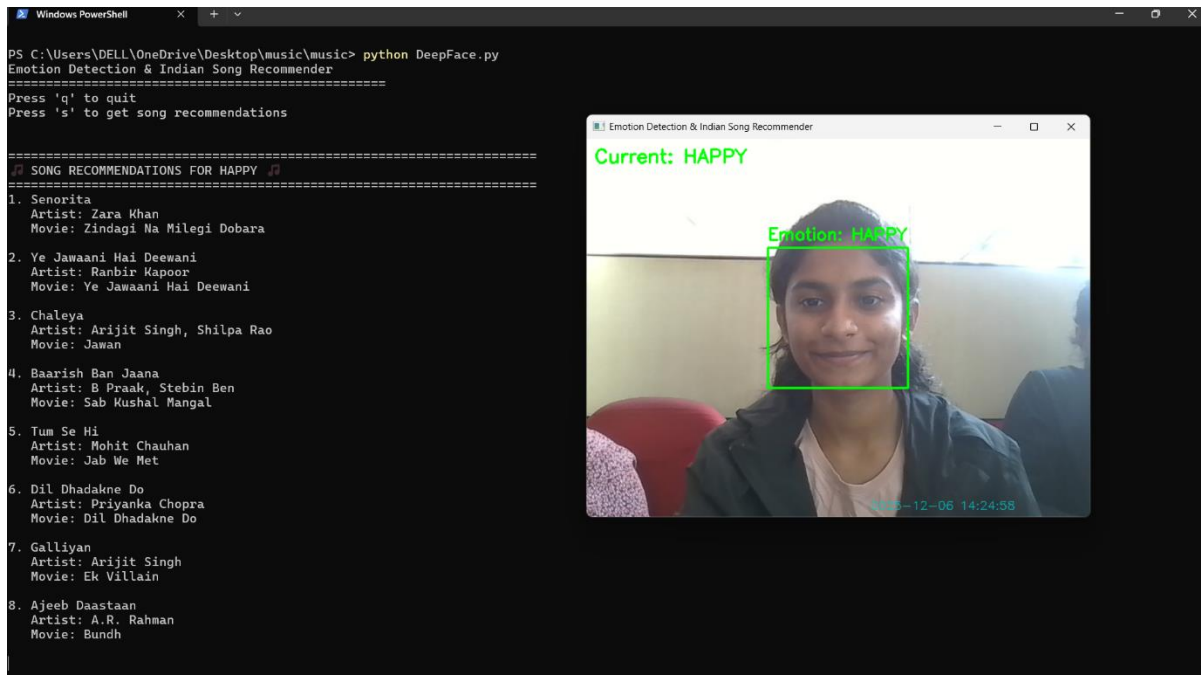
Output: Recommended playlist based on detected emotion

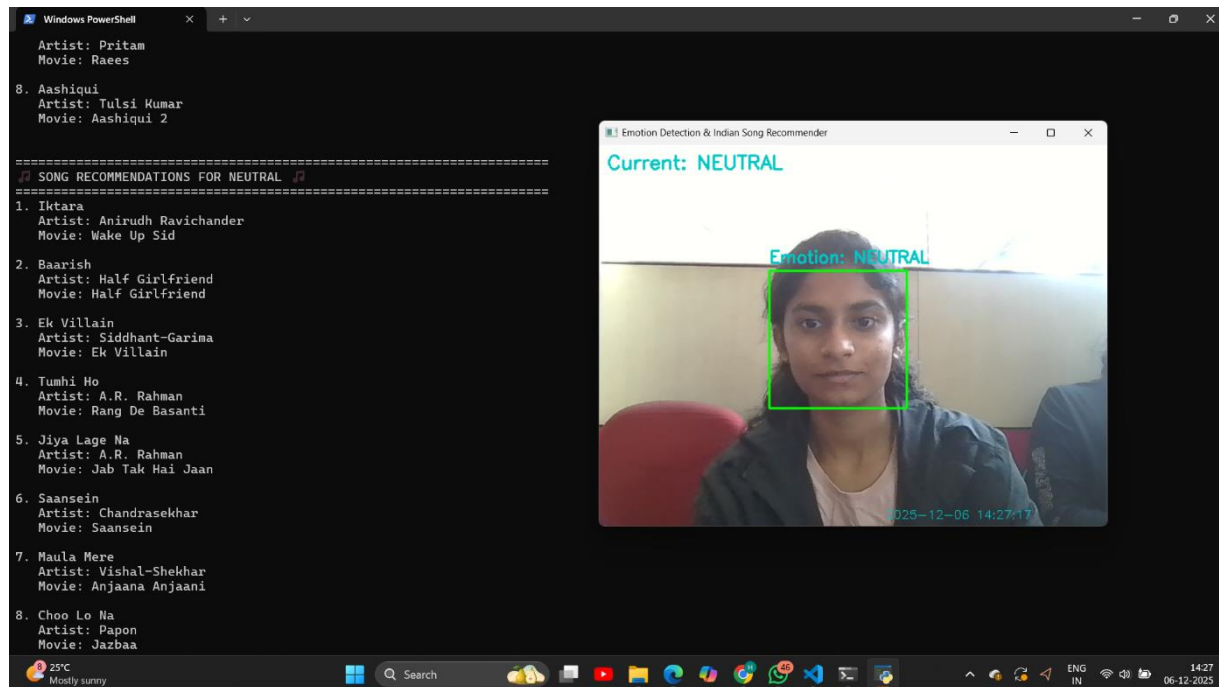
1. Start
2. Capture real-time input from user
  - a. If Image input → capture face image
  - b. If Audio input → capture speech signal
  - c. If Text input → capture typed sentence (optional)
3. Preprocess input data
  - a. Resize image to fixed dimensions
  - b. Convert to grayscale / normalize pixel values
  - c. Remove noise and extract region of interest (face)
  - d. For audio: remove noise, segment signal, extract MFCC
4. Perform Emotion Recognition
  - a. Load trained CNN/LSTM deep learning model
  - b. Feed preprocessed input into model
  - c. Predict emotion label  $E = \{\text{Happy, Sad, Angry, Neutral, etc.}\}$
5. Extract Music Features from database
  - a. Retrieve songs and associated metadata
  - b. Extract audio features (MFCC, tempo, spectral contrast)
  - c. Extract lyrical sentiment (optional)
6. Map detected emotion  $E$  to music attributes  
Example:  
If  $E = \text{Happy}$  → energy=high, tempo=fast  
If  $E = \text{Sad}$  → tempo=slow, acoustic preference, soft tone
7. Compute similarity between emotion vector and song feature vectors  
Use cosine similarity / Euclidean distance
8. Rank songs based on similarity score
9. Select Top-N recommended songs  
Playlist =  $\{S_1, S_2, S_3, \dots, S_n\}$
10. Display recommended playlist on user interface
11. Allow user feedback (like, rating, skip)  
Update recommendation engine with feedback to improve accuracy
12. Stop

## CHAPTER 5

### RESULTS

#### SCREENSHOTS:





## CHAPTER 6

### CONCLUSION

The **Emotion-Driven Music Recommender System** effectively analyzes the user's emotional state and provides personalized music suggestions based on real-time feedback. By combining emotion recognition techniques with intelligent recommendation algorithms, the system enhances the listening experience and delivers music that matches or improves the user's mood. Overall, the project demonstrates how AI and machine learning can be applied to create adaptive, user-centric multimedia applications that support emotional well-being and personalized entertainment.

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